

Problem Statement

Clinical trials frequently suffer from high patient attrition, varying medication adherence rates, and unstratified treatment responses across heterogeneous patient cohorts. In AIDS Clinical Trial Group Study 175, tracking the comparative long-term efficacy of monotherapy versus combination antiretroviral therapies is complicated by patient non-compliance and baseline health variability.

Without centralized data integration and analytics, clinical trial managers struggle to proactively identify drop-out risks, evaluate comparative drug efficacy, and make data-driven protocol adjustments.

Objective & Scope:

To address these challenges, this project leverages Python (Jupyter Notebook), SQL, and Power BI to:

1. Clean and transform clinical trial datasets for longitudinal tracking.
2. Measure treatment efficacy across various drug regimens using baseline and follow-up CD4/health indicators.
3. Calculate medication adherence rates and identify risk factors correlated with non-compliance.
4. Deliver interactive business intelligence dashboards to help clinical trial directors optimize trial protocols, improve patient retention strategies, and accelerate drug efficacy evaluation.

Business Problems Solved using Python

1. **Automated Data Hygiene & Pipeline Validation:**
 - **Problem:** Raw clinical trial data often contains missing values, irregular data types, duplicate patient records, and inconsistent categorical codes that delay analysis and risk regulatory audit failures.
 - **Solution:** Built automated data cleaning, validation, and transformation workflows using Pandas and NumPy to standardize clinical metrics, handle missing records, and generate a clean target dataset (clinical_trial_cleaned.csv) for downstream reporting.
2. **Longitudinal CD4 Trajectory & Biomarker Efficacy Modeling:**
 - **Problem:** Static snapshots fail to capture how patients respond to HIV/AIDS drug regimens over extended trial durations.
 - **Solution:** Processed baseline (cd40), mid-point (cd420), and follow-up (cd496) immune cell counts in Python to compute relative biomarker changes and evaluate the long-term efficacy of monotherapy vs. combination drug regimens.
3. **Demographic & Clinical Stratification (Risk Group Profiling):**
 - **Problem:** Drug safety and efficacy vary across sub-populations, making it difficult to know which patient profiles benefit most from specific drug combinations.
 - **Solution:** Used group-by aggregation and exploratory statistical analysis to segment patients by age, gender, race, hemophilia status, IV drug use history, and prior antiretroviral exposure, identifying sub-groups with higher vulnerability to disease progression.
4. **Treatment Failure & Event Rate Analysis:**
 - **Problem:** Trial sponsors need clear quantitative indicators of clinical failure (e.g.,

disease progression, significant drop in CD4 counts, or mortality) to assess treatment safety profiles.

- **Solution:** Calculated event rates and time-to-event metrics across different drug arms to quantify treatment failure rates, toxicity levels, and overall therapeutic success.

5. **Exploratory Data Analysis (EDA) & Biostatistical Visualization:**

- **Problem:** Raw tabular trial data is difficult for non-technical clinical directors and medical reviewers to digest quickly.
- **Solution:** Generated statistical charts, distribution plots, and correlation matrices using Matplotlib and Seaborn to highlight key relationships between baseline health markers, adherence patterns, and patient outcomes.

Business Problems Solved using SQL

1. **Adherence Threshold & Compliance Rate Segmentation:**

- **Problem:** Trial managers need to separate compliant patients from non-compliant ones based on specific prescription refill/intake thresholds to evaluate whether drug failure is due to low efficacy or poor compliance.
- **Solution:** Used SQL conditional aggregation and window functions to classify patients into adherence brackets (e.g., Medication Possession Ratio / MPR $\geq 80\%$ vs. $< 80\%$), calculating exact adherence percentages across treatment groups.

2. **Longitudinal Patient Health Tracking across Trial Milestones:**

- **Problem:** Clinical data is recorded chronologically across multiple visits, making it difficult to pull baseline vs. post-treatment health status in flat reports.
- **Solution:** Structured SQL queries using JOIN operations, subqueries, and window functions (ROW_NUMBER(), LAG()) to track patient health trajectories (CD4 counts, viral markers) over fixed time intervals.

3. **Cohort Aggregation & Stratified Efficacy Reporting:**

- **Problem:** Biostatisticians need summary statistics (averages, disease progression rates, dropout rates) broken down across multi-variable cohorts (e.g., Treatment Arm + Prior Antiretroviral Exposure).
- **Solution:** Created SQL GROUP BY rollups and CTEs (Common Table Expressions) to compute treatment success rates, average CD4 count improvements, and event rates per demographic segment.

4. **Drop-out & Attrition Rate Identification:**

- **Problem:** Unidentified loss-to-follow-up (dropouts) skews clinical trial results and inflates costs.
- **Solution:** Formulated queries to detect gaps in patient visits and missing follow-up records, isolating high-attrition cohorts for protocol adjustments.

5. **Analytical Data Layer Preparation for BI Dashboards:**

- **Problem:** Visualization tools (like Power BI) perform poorly when fed unorganized raw relational tables.
- **Solution:** Wrote optimized SQL queries and views to transform raw normalized tables into pre-aggregated, analytical data models tailored for direct ingestion into Power BI.

Business Problems Solved using Power BI

1. **Executive Level Clinical Trial Surveillance & Performance Monitoring:**

- **Problem:** Executive sponsors and clinical trial directors struggle to get a unified, high-level view of overall trial health, patient safety, and treatment success metrics from raw datasets.
- **Solution:** Designed high-impact KPI summary cards and executive overview dashboards to track core metrics in real time—such as total enrolled patients, overall retention rates, average CD4 count recovery, and treatment failure rates.

2. **Comparative Drug Arm & Treatment Efficacy Visualization:**

- **Problem:** Biostatisticians need an intuitive way to visually benchmark multi-drug combination regimens against monotherapy regimens without digging through complex statistical output tables.

- **Solution:** Created interactive visual comparisons (bar charts, line graphs, and matrix visuals) to display mean CD4 count changes across different treatment arms over the study duration.
- 3. **Interactive Patient Adherence & Attrition Risk Profiling:**
 - **Problem:** Identifying which specific patient groups suffer from low medication compliance or high dropout rates requires dynamic multi-variable slicing.
 - **Solution:** Built dynamic slicers and cross-filtering visual elements (by age group, gender, race, prior antiretroviral exposure, and IV drug use history) allowing trial managers to drill down into non-compliant cohorts and isolate drop-out drivers.
- 4. **Biomarker Progression & Health Trajectory Dashboards:**
 - **Problem:** Stakeholders need to visually trace patient disease progression over time to evaluate long-term drug sustainability.
 - **Solution:** Visualized longitudinal biomarker trajectories across trial milestones using combo charts and trend lines, highlighting key drop-off points and efficacy plateaus.
- 5. **Self-Service Analytics & Stakeholder Reporting:**
 - **Problem:** Non-technical medical reviewers and operational staff frequently request custom, filtered reports, creating analytical bottlenecks for data teams.
 - **Solution:** Modeled data schemas and implemented DAX measures (calculating custom adherence rates, event frequencies, and percentage changes) to enable self-service exploratory analysis directly through the dashboard interface.